SECURITY INCIDENT RESPONSE

Intrusion Report #2

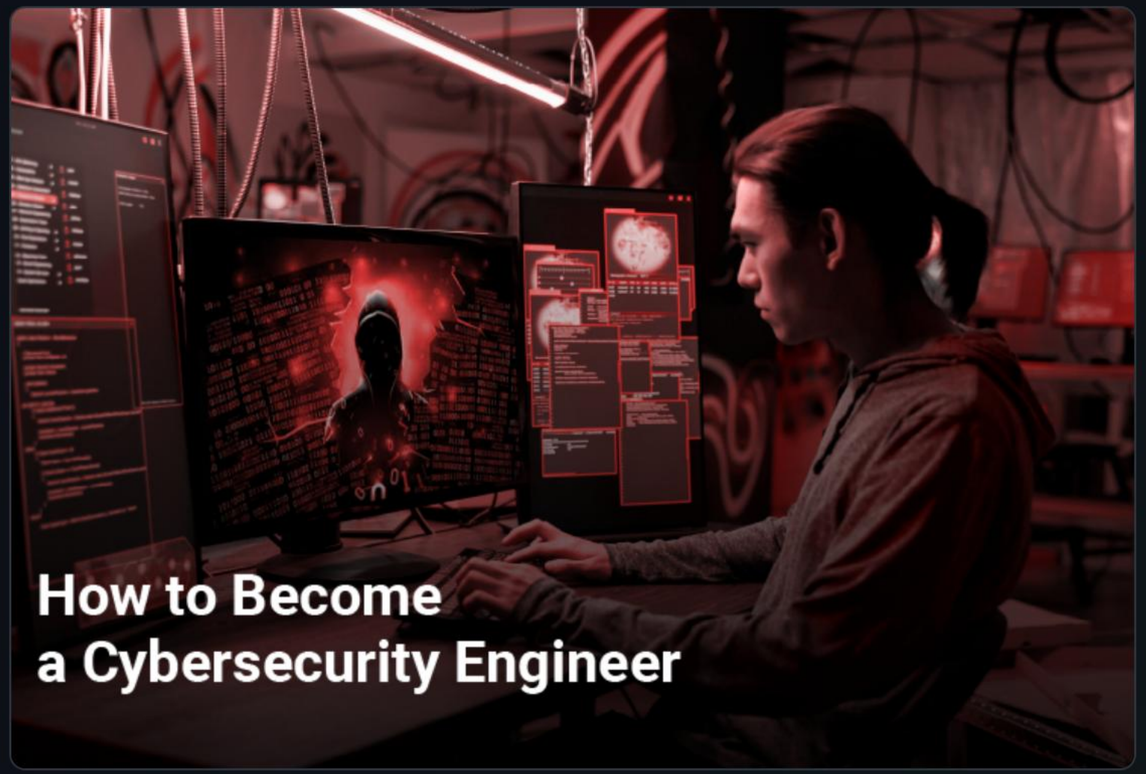
TEAM 40 | ISU CDC | FEBRUARY 21, 2026

STATUS: RESOLVED / MITIGATED

EXECUTIVE SUMMARY

At 1:15 PM, critical infrastructure compromise was detected across the Windows and Linux environments.

- **Unauthorized Access:** Red Team hijacked Domain Controller credentials.
- **Service Sabotage:** MGMT and WWW services rendered unavailable via code removal.
- **Network Hijacking:** Apache process unauthorized binding on Port 80.
- **Mitigation:** 100% of services restored via recovery environment and code repair.



IN-SCOPE INFRASTRUCTURE

AD Server

Windows Active Directory Domain Controller. Central authentication hub for the enterprise environment.

```
ad.team40.isucdc.com
```

MGMT Server

AlmaLinux Management backend. Orchestrates application logic and vehicle telemetry.

```
MGMT_BACKEND / Python
```

WWW Server

Primary Web Front-end. Serves Nginx for public-facing user applications.

```
Port 80 / Nginx
```

INCIDENT TIMELINE

13:15 PM

Initial Domain Controller lockout; NT_STATUS_LOGON_FAILURE detected.

13:30 PM

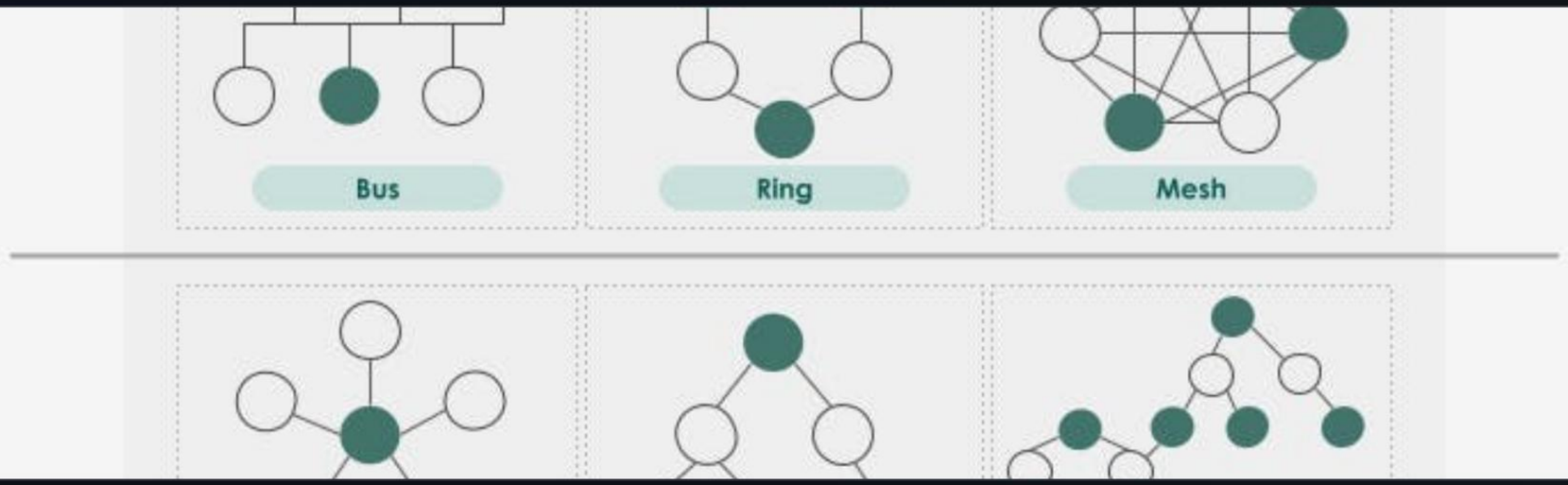
WWW service downtime confirmed; Read timeout on Port 80.

13:45 PM

MGMT backend crash identified as logic sabotage (Missing imports).

14:15 PM

Full service restoration achieved across all target systems.



AD COMPROMISE DETAILS

Credential Hijacking

The Red Team gained system-level access to the Domain Controller, modifying the Administrator password to lock out Team 40. This immediately broke all LDAP-dependent authentication services.

```
ERROR: NT_STATUS_LOGON_FAILURE
Event ID: 4723 - An attempt was made to change an
account's password.
```

Privilege Stripping

Scoring accounts were removed from the **"Transit Admins"** group, preventing the scoring engine from validating service health.

LINUX ENVIRONMENT SABOTAGE

Logic Sabotage

Removal of critical Python imports in MGMT backend caused application-wide NameErrors.

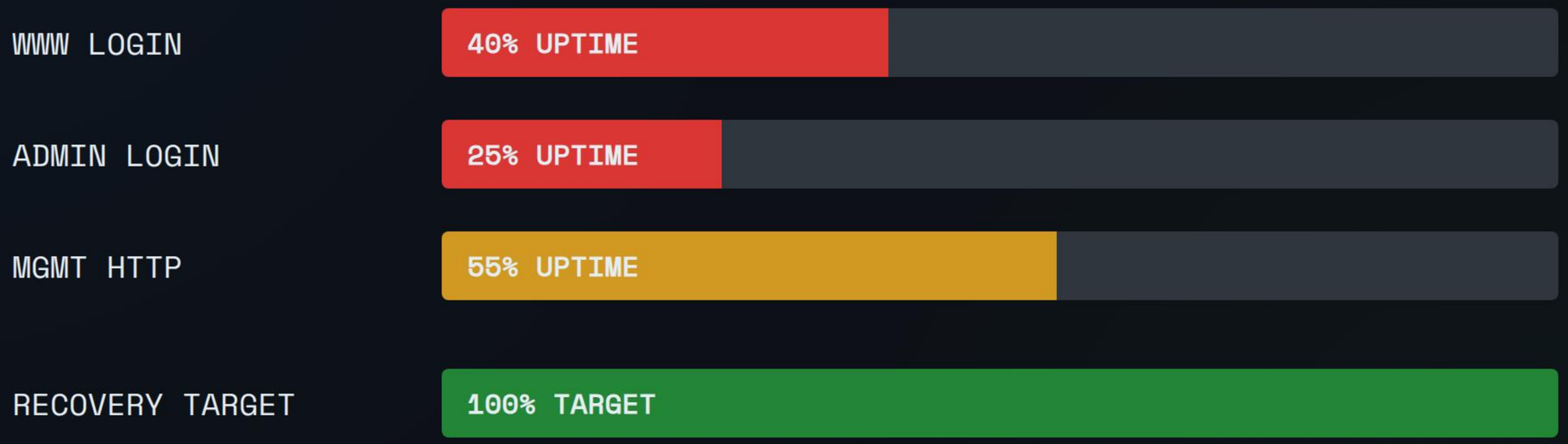
```
# Missing: from flask_login import login_required
```

Port Hijacking

Unauthorized Apache process configured to bind to Port 80, blocking primary Nginx traffic.

```
journalctl: "Address already in use"
```

IMPACT ANALYSIS: SERVICE AVAILABILITY



Downtime resulted in total point loss for multiple scoring intervals during the 13:00-14:00 block.

AD RECLAMATION

Utilman exploit via Recovery ISO

Restoring access required a cold reboot into the Windows Server 2019 ISO environment to manipulate system binaries.

- Booted into Recovery Command Prompt.
- Renamed `utilman.exe` and replaced with `cmd.exe`.
- Triggered admin console from the lock screen.
- Enforced password complexity: `Cdc123456789!`

Mitigation & Recovery

EXECUTING REMEDIATION SCRIPTS...

SYSTEM RECOVERY ACTIONS

System	Mitigation Action	Outcome
AD Domain	Password Reset & Group Membership Restore	SUCCESS
MGMT Backend	Python Import Repair & LDAP Config Update	SUCCESS
WWW Server	Process Termination (fuser -k 80/tcp)	SUCCESS
Auth Flow	Nginx Restart & LDAP_PASS Synchronization	SUCCESS

DEFENSE-IN-DEPTH ROADMAP

- **Multi-Factor Authentication (MFA):** Implement DUO or similar for all Domain Admin logins to prevent credential hijacking.
- **Immutable Auditing:** Configure remote Syslog forwarding to prevent local log tampering by intruders.
- **File Integrity Monitoring:** Deploy Tripwire to alert on changes to `System32` or application config files.
- **Egress Filtering:** Restrict server outbound traffic to block reverse shells and C2 beaconing.

System Status: SECURE

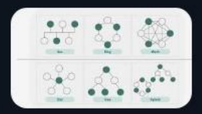
Questions & Debrief

REPORT GENERATED BY TEAM 40 IR UNIT

IMAGE SOURCES

https://eccweb.s3.ap-south-1.amazonaws.com/wp-content/uploads/2023/11/26103754/01-How-to-Become-a-Cybersecurity-Engineer-1.png

Source: www.eccu.edu



https://images.prismic.io/thedecisionlab/aCSbsidWJ-7kSGBb_TDLNetworkTopologyBSRG_Image1.png?auto=format,compress

Source: thedecisionlab.com

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Source: pngtree.com